



Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

**Performance Optimisation of Explainable Machine Learning Models
for Stock Market Crash Prediction**

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This project presents a robust and interpretable machine learning framework for predicting stock market crash events using historical NIFTY 50 data. The proposed methodology employs an optimised soft-voting ensemble model integrating Decision Tree, Logistic Regression, Random Forest, and XGBoost classifiers to enhance predictive performance across varying market regimes. Furthermore, SHAP-based explainable AI techniques are incorporated to ensure model transparency and facilitate the identification of key financial indicators influencing crash prediction. The developed framework contributes towards reliable, data-driven financial risk assessment and supports informed decision-making in volatile market environments.

The Problem — Why Crash Prediction is Challenging

Rare Events / Class Imbalance

- Only ~10.2% of trading days are crashes. Standard ML models are biased toward 'no crash' and miss rare events entirely without special handling.

Non-Linear Market Dynamics

- Markets are driven by panic, herding and structural breaks. Traditional GARCH and rule-based threshold models fail to capture these complex pre-crash patterns.

Black-Box AI Models

- Deep learning models predict well but give no explanation. SEBI guidelines and AI accountability frameworks require transparent, auditable systems.

Major Crashes Studied

- Dot-com (2001-02) | GFC (2008-09) | European Debt Crisis (2011) | COVID-19 (2020) | Inflation Shock (2022)

Our Solution — XAI-Augmented Ensemble Framework

01 — Soft-Voting Ensemble

- Combines Decision Tree, Logistic Regression, Random Forest and XGBoost via soft-voting probabilities for optimised ROC-AUC performance.

02 — SHAP Explainability

- SHapley Additive exPlanations (TreeExplainer) decompose every prediction into per-feature contributions — fully transparent and auditable.

03 — SMOTE Oversampling

- Synthetic Minority Over-sampling balances the severe 10.2% crash class imbalance, ensuring the model learns crash patterns effectively.

Key Results

- ROC-AUC (Ensemble): 0.627 | +7.5% vs Decision Tree | RF Crash Recall: 79.2% | 6,211 Trading Days Tested

Traditional Methods — Rule-Based and Econometric

- VIX fear gauge and yield curve inversions generate excessive false positives and fail to model multi-feature interactions in non-stationary markets.

Deep Learning — LSTM-Based Predictions

- LSTMs outperform classical models but are black-boxes. Only 12% of 150 studies addressed interpretability. ROC-AUC 0.85 achieved without explainability.

Explainable AI — SHAP and XAI Methods

- SHAP is the most theoretically grounded post-hoc method. XGBoost+SHAP applied to systemic risk. Accuracy-interpretability trade-off remains open.

Ensemble Methods — Crisis Prediction

- Ensemble methods consistently outperform single estimators in financial crisis prediction, motivating this study's soft-voting approach.

Research Gap

- No prior work applies soft-voting ensemble with SHAP to NIFTY 50 crash prediction with temporal rolling-window validation.

NIFTY 50 Index — NSE India

- Period: January 2001 – March 2026 | Total Observations: 6,260 daily records → 6,211 clean
- Features: Open, High, Low, Close (OHLC) + Volume | Crash Rate: ~10.2% of observations
- Crash Label: 5-day forward return $< -3\%$ → $y=1$ | Train: 2001–2019 (4,675 samples) | Test: 2020–2023 (994 samples)

25 Engineered Features

- Price Returns: 1/3/5/10/20-day returns | Moving Averages: MA deviation, MA cross signals
- Volatility: Rolling vol (5/10/20-day), ATR 14-day | Momentum: RSI, MACD, Histogram, Stochastic
- Bollinger Bands: Band width, price position | Volume: Volume ratio, OBV (normalised)
- Candlestick: High-Low range relative to close

Crash Episodes Covered

- Dot-com | GFC | European Debt Crisis | COVID-19 | Inflation Shock

Methodology — End-to-End Pipeline

Step 01 — Data Collection

- NIFTY 50 daily OHLCV from NSE, covering 2001–2026 with 6,211 records.

Step 02 — Feature Engineering

- 25 technical and volatility features computed via rolling window analysis.

Step 03 — Crash Labelling

- 5-day forward return $< -3\%$ defines crash. SMOTE applied to handle 10.2% crash class imbalance.

Step 04 — Model Training

- Models trained: Decision Tree, Logistic Regression, Random Forest, XGBoost, Soft-Voting Ensemble.

Step 05 — SHAP and Validation

- TreeExplainer for SHAP values. ROC-AUC evaluation with 10-quarter rolling-window temporal validation.

Output Metrics

- ROC-AUC | Precision / Recall / F1 | SHAP Feature Importance | Confusion Matrix | PR Curves | Ablation Study

Baseline — Decision Tree

- Max depth=6, CART algorithm. Simple, fully interpretable. Highest variability across windows.

Baseline — Logistic Regression

- L2-regularized, C=0.5, StandardScaler. Linear probabilistic baseline. Lowest FP among high-recall models (206 FP).

Strong — Random Forest

- 200 trees, Max depth=8, Bootstrap. Highest recall 79.2% — captures non-linear crash dynamics via bagging.

Strong — XGBoost

- 200 estimators, LR 0.05, Subsample 0.80. Gradient boosting. Best accuracy (0.663). Complements RF in ensemble.

Our Model — Soft-Voting Ensemble (Best ROC-AUC 0.627)

- LR + RF + XGBoost with soft probabilities. Balances precision, recall and calibration.
- All models: SMOTE-balanced | StandardScaler on LR and Ensemble | Train: 2001–2019 | Test: 2020–2023

Results — Model Performance on Test Set (2020–2023)

Performance Summary

- Decision Tree: Accuracy 0.498 | Recall 0.740 | ROC-AUC 0.583
- Logistic Regression: Accuracy 0.619 | Recall 0.571 | ROC-AUC 0.579
- Random Forest: Accuracy 0.489 | Recall 0.792 | ROC-AUC 0.634
- XGBoost: Accuracy 0.663 | Recall 0.494 | ROC-AUC 0.613
- Ensemble (Ours): Accuracy 0.597 | Recall 0.610 | ROC-AUC 0.627

Key Highlights

- Ensemble ROC-AUC: 0.627 — +7.5% improvement over Decision Tree baseline.
- Random Forest Recall: 79.2% — highest crash detection rate across all models.
- Test Set: 994 samples including 77 crash events (7.7% crash prevalence).
- Primary metric is ROC-AUC — unaffected by class imbalance. All models exceed random baseline (0.500).

ROC Curve Analysis — All Five Models

AUC Scores

- Random Guess Baseline: AUC 0.500 | Logistic Regression: AUC 0.579
- Decision Tree: AUC 0.583 | XGBoost: AUC 0.613
- Random Forest: AUC 0.634 | Ensemble (Best Overall): AUC 0.627

Key Findings

- Ensemble wins overall — AUC 0.627 provides best balance of precision and recall across all thresholds.
- Random Forest achieves best recall — AUC 0.634, captures 79.2% of crashes. Most aggressive at high recall.
- XGBoost converges at high FPR — AUC 0.613, outperformed by ensemble at higher false positive rates.
- All 5 models beat the random baseline — demonstrating meaningful predictive signal in the feature set.

Evaluation Note

- ROC-AUC is the primary metric as it is threshold-independent and robust to class imbalance.

SHAP Feature Importance — Top 10 Predictors

Top Volatility Features (Highest SHAP Values)

- 1. ATR (14-day): 0.0502 | 2. Volatility (20-day): 0.0422 | 3. High-Low Range: 0.0277
- 4. Volatility (10-day): 0.0213 | 5. MA20-MA50 Cross: 0.0185
- 6. MACD Histogram: 0.0159 | 7. MA5-MA20 Cross: 0.0137
- 8. Volatility (5-day): 0.0124 | 9. 20-day Return: 0.0122 | 10. MACD Signal: 0.0109

Key Insights

- ATR Dominates: Price range expansion is the #1 crash precursor — confirms volatility expansion precedes crashes.
- ~34% Attribution: Top 3 volatility features account for 34% of all SHAP attribution.
- Theory Confirmed: SHAP rankings align with classical financial theory on tail-risk indicators.
- SEBI Compliant: Per-prediction explanations enable regulatory-grade audit trails for deployment.

Temporal Validation — Rolling-Window Analysis

Validation Setup

- Window Size: 3 months (non-overlapping) | Period: 2020–2023 | Valid Windows: 10 quarters
- Metric: F1-score per window per model | Purpose: Test temporal stability and regime robustness

Q1 2020 — COVID-19 Liquidity Shock

- High volatility spike. Most models detect crashes well. Decision Tree achieves peak F1 but with high variance.

2020–21 — Recovery and Normalisation

- Models show distributional drift. Performance drops as market stabilises — crash patterns become sparse.

2022 — Inflation / Rate-Hike Shock

- Ensemble and RF recover predictive power. Structural shift from pandemic regime confirmed by F1 variance.

Average F1 Across 10 Windows

- DT: 0.186 | LR: 0.188 | RF: 0.194 | XGBoost: 0.185 | Ensemble: 0.190

Confusion Matrix Results (F1-Optimal Thresholds)

- Decision Tree: TP=35 FP=386 TN=531 FN=42
- Logistic Regression: TP=41 FP=206 TN=711 FN=36
- Random Forest: TP=67 FP=592 TN=325 FN=10
- XGBoost: TP=50 FP=404 TN=513 FN=27
- Ensemble (Ours): TP=66 FP=497 TN=420 FN=11

Key Findings

- Ensemble Best Balance: TP=66, FP=497 vs RF's FP=592. Soft-voting smooths extreme FP clusters while maintaining near-identical TP count.
 - Random Forest Best TP: TP=67 (87% of crashes detected). Highest recall but costliest in false positives (592 false alarms).
 - LR Fewest False Positives: Only 206 FP — most precision-efficient baseline for low-tolerance deployments.
- Test Set: 994 samples | 77 crash events (7.7%)

Average Precision (AP) Scores

- No-Skill Baseline: $AP=0.077$ | Decision Tree: $AP=0.077$
- Random Forest: $AP=0.094$ | XGBoost: $AP=0.102$
- Ensemble: $AP=0.112$ | Logistic Regression (Best): $AP=0.146$

PR Curves vs ROC Curves

- For imbalanced datasets (7.7% crash rate), PR curves are more discriminative — avoids inflation from large true negative counts.

Key Findings

- LR Highest $AP=0.146$: Superior precision-recall tradeoff despite lower ROC-AUC.
 - Ensemble $AP=0.112$: Strong AP across full recall spectrum (0–1.0), allowing practitioners to choose operating points by risk tolerance.
 - All Models Beat No-Skill Baseline (0.077): Confirms genuine discriminative ability beyond random chance.
- Dashed line = no-skill baseline ($AP = \text{crash prevalence} = 0.077$)

Key Findings

- Soft-voting ensemble achieves ROC-AUC 0.627 on 2020–2023 NIFTY 50 test set — 7.5% above Decision Tree baseline.
- SHAP identifies ATR (14-day), 20-day Volatility and High-Low Range as top 3 crash precursors — aligned with financial theory.
- Temporal robustness validated across COVID-19, inflationary shock and post-pandemic recovery regimes via 10-quarter rolling windows.
- XAI-augmented predictions provide SEBI-compatible, auditable explanations — bridging accuracy and regulatory transparency.

Limitations

- Price-only features (no macroeconomic or sentiment data) | Static crash threshold (-3% / 5-day) | Single index (NIFTY 50)

Future Work

- Dynamic crash thresholds via rolling volatility z-scores.
- Macroeconomic features and NLP sentiment signals for enhanced prediction.
- Transfer learning from developed markets to improve generalisation.
- LIME + SHAP cross-validation for attribution robustness.



Thank You